SURYAGRID AI

Grid Operations RFP and Evaluation Brief

Business objective, demand/reserve context, RFP questions and scoring framework

Document Type	Grid / RFP Evaluation Brief
Version	v1.0
Use	Sample PDF upload for LLM requirement extraction, planning and forecast design
Data Status	Synthetic demonstration - not real plant SCADA or commercial data
Primary Objective	Enable a model to infer business use case, horizon, variables, methods, metrics and data gaps

LLM extraction note: This document intentionally uses clear labels, tables and repeated field names so a document-reading model can extract business objectives, SCADA/MET/NWP variables, forecast horizons, metrics and RFP requirements with minimal ambiguity.

1. Objective of This Brief

This brief shows how a forecasting RFP should connect business objectives to forecast horizons, required data, methodology and evaluation metrics. It is intended for upload into the LLM planner so the model can generate a complete solution recommendation instead of a generic forecast answer.

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LLM-readable context: DOCUMENT_TYPE = GRID_OPERATIONS_RFP_REQUIREMENT_BRIEF. KEY_REQUIREMENT = business objective
first, then horizon, variables, method, metrics and acceptance tests. INCLUDE = demand_forecast_mw and
reserve_requirement_mw when objective includes grid balancing or dynamic reserve.
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2. Business Objective to Data Requirement Map

Business objective	Operational question	Required external/business variables	Forecast outputs required
Deviation penalty reduction	How far will actual output deviate from schedule?	scheduled_mw, dispatch_block_id, deviation_penalty_rate	P50 forecast, error estimate, deviation risk flag
Grid dispatch support	What renewable output should the grid expect?	demand_forecast_mw, reserve_requirement_mw, grid_constraint_flag	P10/P50/P90, ramp risk, curtailment flag
Dynamic reserve calculation	How much reserve is needed for renewable uncertainty?	demand_forecast_mw, reserve_requirement_mw, NWP ensemble spread	low-generation risk, uncertainty band, reserve contribution
Critical ramp warning	Will output change fast enough to require action?	ramp_rate_mw_min, cloud_obstruction_probability, SCADA freshness	event probability, lead time, phase error
Market revenue optimization	How do forecast uncertainty and prices affect dispatch value?	market_price_forecast, deviation_penalty_rate, schedule constraints	scenario revenue range and risk-adjusted schedule

3. RFP Questions Vendors Must Answer

RFP question	Why it matters
Which forecast horizons are supported, and which business decisions does each horizon serve?	Prevents vendor from offering one generic forecast without operational context.
What SCADA, meteorological, NWP and market/grid variables are mandatory before go-live?	Identifies data dependencies and integration gaps.
Which method family is used by horizon: persistence, physical model, ensemble, analog, calibration?	Method must match horizon and decision.
How are uncertainty bands generated and calibrated?	Reserve and day-ahead decisions need probability, reliability and coverage.
How are curtailment, stale SCADA and equipment derating separated from weather forecast error?	Avoids incorrect model blame and poor operator trust.
How will performance be evaluated during ramp events, curtailment windows and normal days?	Average error alone hides event failures.

4. Proposal Scoring Matrix

Evaluation area	Weight	Scoring guide
Business objective alignment	20%	Forecast output must explicitly support deviation, reserve, dispatch and ramp decisions.
Data interface completeness	20%	Supports SCADA, met, NWP, grid/demand and event labels with quality rules.
Methodology by horizon	20%	Different horizons use appropriate model families and uncertainty methods.
Probabilistic validation	15%	Shows P10/P90 coverage, reliability and rank histogram for day-ahead/weekly.
Ramp and event validation	15%	Shows ramp hit rate, lead time, false alarm, phase and amplitude error.
Explainability and operator workflow	10%	Provides plain-language action, root cause, hierarchy drilldown and audit trail.

5. Minimum RFP Requirement Statements

- Vendor shall provide horizon-wise method description, not a single generic model statement.

- Vendor shall accept SCADA actual, possible power, schedule, local limit, availability and data quality fields.
- Vendor shall accept NWP issue_time, valid_time and ensemble member identifiers for day-ahead probabilistic forecasting.
- Vendor shall provide P10, P50 and P90 outputs where uncertainty-aware decisions are required.
- Vendor shall validate forecasts using separate normal-day, ramp-day, curtailment and stale-data subsets.
- Vendor shall include demand_forecast_mw and reserve_requirement_mw fields when the objective includes grid balancing or dynamic reserve.
- Vendor shall report missing data impact and indicate which horizons are enabled, partially enabled or not enabled.

6. Sample Response Format Expected from LLM


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{"business_use_case":"dynamic_reserve_calculation","recommended_horizons":["2hour","4hour","day_ahead_40hour"],"mandatory_variables":["actual_mw","possible_power_mw","nwp_ensemble_member_id","demand_forecast_mw","reserve_requirement_mw"],"method_family":"nwp_ensemble_quantile_calibration","metrics":["coverage","reliability","rmse","bias","crps_proxy"],"missing_information":["exact reserve policy"]}
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7. Clarification Questions for User

- Is the primary business target deviation penalty reduction, dynamic reserve, market revenue or all three?
- What is the dispatch block length and penalty formula?
- Do forecasts need to be delivered at grid, plant, hub and block level?
- Which NWP forecast providers and issue cycles are available?
- Is curtailment ground truth available or must it be inferred from local_limit_mw and possible_power_mw?
- Should day-ahead output be delivered as P10/P50/P90 only or full ensemble scenarios?